Ex.8 : UDP-based DOB exchange and Age calculation

1. The client asks the user to enter their date of birth and sends it to the server.
2. The server receives the DOB, calculates the age, and sends it back to the client.
3. The client receives the age and prints it to the console.

UDPClient.java

```
import java.io.*;
import java.net.*;
import java.time.*;
import java.time.format.DateTimeFormatter;
import java.util.*;

public class UDPClient {
   public static void main(String[] args) throws IOException {
      DatagramSocket socket = new DatagramSocket();
      InetAddress address = InetAddress.getLocalHost();
      int port = 8000;

      Scanner scanner = new Scanner(System.in);
      System.out.println("Enter your date of birth (yyyy-MM-dd): ");
      String dob = scanner.nextLine();

      // Send DOB to server
      byte[] data = dob.getBytes();
      DatagramPacket packet = new DatagramPacket(data, data.length, address, port);
      socket.send(packet);

      // Receive age from server
      packet = new DatagramPacket(new byte[1024], 1024);
      socket.receive(packet);
      String age = new String(packet.getData(), 0, packet.getLength());
      System.out.println("Your age is: " + age);

      socket.close();
   }
}
```

UDPServer.java

```java
import java.io.*;
import java.net.*;
import java.time.*;
import java.time.format.DateTimeFormatter;
import java.util.*;

public class UDPServer {
    public static void main(String[] args) throws IOException {
        DatagramSocket socket = new DatagramSocket(8000);
        System.out.println("Server started. Waiting for client...");

        while (true) {
            DatagramPacket packet = new DatagramPacket(new byte[1024], 1024);
            socket.receive(packet);

            String dob = new String(packet.getData(), 0, packet.getLength());
            System.out.println("Received DOB: " + dob);

            String age = calculateAge(dob);

            // Send age to client
            byte[] data = age.getBytes();
            InetAddress address = packet.getAddress();
            int port = packet.getPort();
            DatagramPacket response = new DatagramPacket(data, data.length, address, port);
            socket.send(response);
        }
    }

    public static String calculateAge(String dob) {
        DateTimeFormatter formatter = DateTimeFormatter.ofPattern("yyyy-MM-dd");
        LocalDate birthDate = LocalDate.parse(dob, formatter);
        LocalDate currentDate = LocalDate.now();
        Period period = Period.between(birthDate, currentDate);
        int years = period.getYears();
        int months = period.getMonths();
        int days = period.getDays();
        return years + " years, " + months + " months, " + days + " days";
    }
}
```

Server Output:

```
Server started. Waiting for client...
Received DOB: 2006-01-01
```

Client Output:

```
Enter your date of birth (yyyy-MM-dd):
2006-01-01
Your age is: 20 years, 1 months, 2 days
```